Lecture 24

The Lax pair of KdV

Lecture 23 obtained KdV from a Hamiltonian. The same equation can be written as the evolution of a differential operator. We compute the commutator that gives this description and specify the solution class used in the following scattering lectures.

Objectives. *We prove the Lax identity by applying both compositions to a test function. We state the cited existence and decay results that supply smooth Schwartz solutions. We distinguish this differential identity from a spectral theorem.*

By the end of this lecture you will be able to:

- Expand the ordered commutator and cancel all differentiated test terms.
- Recover KdV from equality on every compactly supported smooth test.
- State the decay hypothesis and the limit of the formal operator argument.

24.1 Differential operators on test functions

Let $I \subset \mathbb{R}$ be an open interval and let $u : I \times \mathbb{R} \to \mathbb{R}$ be smooth. Write $D = \partial_x$. A test function here belongs to $C_c^\infty(\mathbb{R}, \mathbb{C})$, the smooth complex functions with compact support.

Definition 24.1 (Lax representation)**.** The *commutator* of differential operators is $[B, L] = B \circ L - L \circ B$. A *Lax representation* of an evolution equation is an identity $L_t = [B, L]$ equivalent to that equation. For KdV the proposed operators are

$$L = -D^2 + u, \qquad B = -4D^3 + 3(uD + Du). \tag{24.1}$$

Here u denotes multiplication, and Du means $D \circ (u\,\cdot)$. The time derivative L_t differentiates the coefficients with the test function held fixed.

Both operators map C_c^∞ into itself, so both compositions have a common domain. The product rule gives

$$B\psi = -4\psi''' + 6u\psi' + 3u_x\psi, \qquad L_t\psi = u_t\psi. \tag{24.2}$$

Thus the term Du in (24.1) includes both $u\psi'$ and $u_x\psi$. Omitting either contribution changes the equation. Primes on a test function denote spatial derivatives.

24.2 The commutator gives KdV

Proposition 24.2 (KdV Lax identity). *For smooth real u on $I \times \mathbb{R}$, the identity*

$$L_t\psi = [B, L]\psi \qquad \text{for every } \psi \in C_c^\infty(\mathbb{R}, \mathbb{C}),\ t \in I$$

holds if and only if

$$u_t - 6uu_x + u_{xxx} = 0 \quad \text{on } I \times \mathbb{R}. \tag{24.3}$$

More precisely, $[B, L]$ is multiplication by $6uu_x - u_{xxx}$ on these tests.

Proof. At a fixed time, distribute the commutator over $L = -D^2 + u$ and $B = -4D^3 + 6uD + 3u_x$. Constant-coefficient derivatives commute. For the other terms, repeated product differentiation gives

$$\begin{aligned} [-4D^3, u]\psi &= -4u_{xxx}\psi - 12u_{xx}\psi' - 12u_x\psi'', \\ [6uD, -D^2]\psi &= 6u_{xx}\psi' + 12u_x\psi'', \\ [3u_x, -D^2]\psi &= 3u_{xxx}\psi + 6u_{xx}\psi', \\ [6uD, u]\psi &= 6uu_x\psi, \qquad [3u_x, u]\psi = 0. \end{aligned} \tag{24.4}$$

For example, the first line uses $(u\psi)''' = u_{xxx}\psi + 3u_{xx}\psi' + 3u_x\psi'' + u\psi'''$; the last summand cancels the other ordered composition. The second line follows by expanding $(u\psi')''$. Adding the displayed terms gives

$$[B, L]\psi = (6uu_x - u_{xxx})\psi + (-12 + 6 + 6)u_{xx}\psi' + (-12 + 12)u_x\psi''.$$

Both derivative coefficients vanish. This proves the asserted multiplication formula.

If u satisfies KdV, this formula equals $u_t\psi = L_t\psi$. Conversely, the assumed operator equality gives $(u_t - 6uu_x + u_{xxx})\psi = 0$ for every test. At any chosen (t, x_*), select a test equal to one near x_*. The coefficient must vanish at that point. The point was arbitrary, so KdV holds everywhere. No spatial decay is required for this proof. ∎

The ordering is part of the statement. Replacing $[B, L]$ by $[L, B]$ negates its scalar coefficient. Dunajski, *Solitons, Instantons, and Twistors*, §2.2.4, pp. 31–32, writes $L_t = [L, A]$ with $A = -B$; that convention gives the same identity.

Example 24.3 (Determining the coefficients). Start with $B_{r_3,r_1,r_0} = r_3D^3 + r_1uD + r_0u_x$, where r_3, r_1 and r_0 are real constants. The same product rule gives

$$\begin{aligned} [B_{r_3,r_1,r_0}, L]\psi = (3r_3 + 2r_1)u_x\psi'' + (3r_3 + r_1 + 2r_0)u_{xx}\psi' \\ + \left[(r_3 + r_0)u_{xxx} + r_1uu_x\right]\psi. \end{aligned}$$

Cancellation for arbitrary smooth u forces $r_1 = -3r_3/2$ and $r_0 = -3r_3/4$. Choosing $r_3 = -4$ fixes $r_1 = 6$ and $r_0 = 3$. The remaining coefficient is then exactly $6uu_x - u_{xxx}$. This calculation determines the relative coefficients rather than assuming them.

24.3 The standing Schwartz solution class

Recall the real Schwartz space $\mathcal{S} = \mathcal{S}(\mathbb{R}, \mathbb{R})$ from Lecture 23. Its seminorms are $p_{N,j}(f) = \sup_x (1 + |x|)^N |\partial_x^j f(x)|$ for integers $N, j \geq 0$. For the time-dependent scattering statements in Lectures 24–26, use

$$u \in C^\infty(\mathbb{R}; \mathcal{S}), \qquad u_t - 6uu_x + u_{xxx} = 0. \tag{24.5}$$

Smoothness here is in the Schwartz topology. In particular, $\sup_{t\in K} p_{N,j}(u(t)) < \infty$ for every compact time interval $K \subset \mathbb{R}$ and every N, j. This is stronger than membership in $\mathcal{S}$ at each separate time. For $w = \partial_t^k u$, continuity into $\mathcal{S}$ gives

$$|\partial_x^j w(t,x) - \partial_x^j w(s,y)| \le p_{0,j}(w(t) - w(s)) + |\partial_x^j w(s,x) - \partial_x^j w(s,y)|.$$

The first term tends to zero as $t \to s$, uniformly in x; the second tends to zero as $x \to y$. Thus all mixed derivatives are jointly continuous, and the field is jointly smooth in (t,x), so Proposition 24.2 applies at every time.

Existence for arbitrary real $u_0 \in \mathcal{S}$ in this class is a *cited consequence* of Sobolev existence and weighted persistence. The relevant statements, in the source convention $v_t + vv_x + v_{xxx} = 0$, are the following.

Bona–Scott, *Solutions of the Korteweg–de Vries equation in fractional order Sobolev spaces*, Duke Math. J. 43 (1976), Theorem 3, p. 95, gives the global solution map in $H^s(\mathbb{R})$ for every $s \ge 2$. The map takes values continuously in H^s on each finite time interval. Here $\|f\|_{H^s}^2 = \int (1+k^2)^s |\widehat{f}(k)|^2 \mathrm{d}k$, with the Fourier convention of Lecture 4. Uniqueness identifies the solutions at different regularities, so Schwartz data give one solution in every such Sobolev space. The source treats positive times. To obtain negative times for datum v_0, solve forward from $v_0(-x)$ and reflect that solution by $(t,x) \mapsto (-t,-x)$. Its value at zero is then $v_0(x)$.

Nahas–Ponce, *On the persistence properties of solutions of nonlinear dispersive equations in weighted Sobolev spaces*, Theorem 2, author PDF p. 6, proves that a Sobolev solution continuous in H^m, $m \ge 1$, retains continuous membership in $L^2(|x|^m \mathrm{d}x)$ on the same time interval whenever its initial datum has that weight. This theorem includes KdV. The smooth Sobolev solutions above agree with the local solution to which the persistence theorem applies, by uniqueness. Using every large integer m therefore supplies every polynomial L^2 weight on every finite interval. Finally set $v = -6u$ to obtain exactly (24.3).

We spell out why these cited statements give the uniform Schwartz hypothesis. Put $\langle x \rangle = (1+x^2)^{1/2}$. Lemma 3 of Nahas–Ponce, author PDF p. 5, gives, for $N, n \ge 0$ integers,

$$\|\langle x\rangle^{N+1} f\|_{H^{n+1}} \le C_{N,n} \|\langle x\rangle^{2N+2} f\|_2^{1/2} \|f\|_{H^{2n+2}}^{1/2}. \tag{24.6}$$

Indeed, take its parameters $a = 2n+2$, $b = 2N+2$ and $\theta = 1/2$. Leibniz's rule and the one-dimensional embedding $H^1 \subset C_b^0$ then give

$$p_{N,n}(f) \le C'_{N,n} \|\langle x\rangle^{N+1} f\|_{H^{n+1}}.$$

To see the derivative step, write $f = \langle x\rangle^{-N-1}(\langle x\rangle^{N+1} f)$ and differentiate n times. Multiplication by $(1+|x|)^N$ leaves bounded coefficients. Every derivative of the parenthesised function through order n is bounded by its H^{n+1} norm.

Apply these estimates to $f = u(t) - u(s)$. Weighted persistence with $m = 4N+4$ makes the weighted norm on the right of (24.6) tend to zero as $t \to s$. Sobolev continuity controls the other factor. Thus u is continuous in every Schwartz seminorm, uniformly bounded on compact time intervals. The map $f \mapsto 6ff' - f'''$ is a continuous polynomial map $\mathcal{S} \to \mathcal{S}$. The PDE first gives $u \in C^1(\mathbb{R}; \mathcal{S})$: integrate this continuous right-hand side in the complete Schwartz space and use the known Sobolev equation to identify the integral with $u(t) - u(s)$. Repeated differentiation of the polynomial right-hand side then gives C^∞ time dependence. This derives (24.5); it does not reprove the cited Cauchy and persistence theorems.

Example 24.4 (The Lax derivative at a Gaussian datum)**.** Take $u_0(x) = e^{-x^2}$. The cited result supplies a solution through this datum. Its initial time derivative follows without finding that solution:

$$\begin{aligned} u_t(0,x) &= 6u_0u_0' - u_0''' \\ &= (8x^3 - 12x)e^{-x^2} - 12xe^{-2x^2}. \end{aligned}$$

At $x = 1$ the value is $-4e^{-1} - 12e^{-2} \neq 0$. For a test equal to one near this point, the value of $[B(0), L(0)]\psi$ is the same number. The reversed commutator has the opposite value. This example tests both terms in the nonlinear evolution.

24.4 The boundary of the operator statement

The test-function calculation also gives formal symmetry properties. For compactly supported smooth complex f and g, integration by parts uses the endpoint primitives

$$\begin{aligned} \overline{f}Lg - \overline{Lf}g &= D(\overline{f'}g - \overline{f}g'), \\ \overline{f}Bg + \overline{Bf}g &= D\Big[-4(\overline{f}g'' - \overline{f'}g' + \overline{f''}g) + 6u\overline{f}g\Big]. \end{aligned}$$

Both primitives vanish outside a compact set. Hence $(f, Lg) = (Lf, g)$ and $(f, Bg) = -(Bf, g)$ on this test domain, where the L^2 inner product is antilinear in its first slot. These identities mean that L is *formally symmetric* and B is *formally skew-symmetric* there.

They do not identify closed operator domains or construct an evolution operator on L^2. The operators contain derivatives and are unbounded on L^2. A finite-matrix argument that solves $U_t = BU$ and writes $L(t) = U(t)L(0)U(t)^{-1}$ needs additional analytic results in this setting. Proposition 24.2 proves equality of differential operators on tests. It makes no claim that the spectrum stays fixed. Lecture 26 treats the time evolution of bound-state data under its stated additional differentiability hypothesis. Lecture 25 first constructs the scattering solutions for each fixed real Schwartz potential.

Summary. For $L = -\partial_x^2 + u$ and $B = -4\partial_x^3 + 6u\partial_x + 3u_x$, the commutator on compactly supported smooth tests is multiplication by $6uu_x - u_{xxx}$. Thus $L_t = [B, L]$ is equivalent to KdV for smooth u. Cited Sobolev existence and weighted persistence, together with the displayed interpolation estimate, supply $C^\infty(\mathbb{R}; \mathcal{S})$ solutions for real Schwartz data. The test-function identity alone is not an isospectrality theorem.

Self-check. What does Du mean in $B = -4D^3 + 3(uD + Du)$? Compute its action on a test.

Answer. It means differentiation after multiplication: $(Du)\psi = (u\psi)'$. Thus $B\psi = -4\psi''' + 6u\psi' + 3u_x\psi$. Replacing Du by multiplication by u_x loses one $3u\psi'$ term.

Self-check. Which two coefficients in the commutator cancel the differentiated tests? What remains when the order of the commutator is reversed?

Answer. The coefficients are $(-12 + 12)u_x$ for ψ'' and $(-12 + 6 + 6)u_{xx}$ for ψ'. The reversed order leaves multiplication by $u_{xxx} - 6uu_x$. It therefore cannot replace $[B, L]$ with L_t unchanged.

Self-check. Why does Schwartz membership at every separate time not state the full standing hypothesis? What supplies the stronger conclusion here?

Answer. Separate membership does not assert continuity or uniform compact-time bounds in any Schwartz seminorm. Cited continuous Sobolev and weighted norms, applied to time differences in (24.6), give that continuity. Compactness then gives the uniform bounds.

Self-check. Does the proved Lax identity establish constancy of every eigenvalue of an unbounded operator on L^2?

Answer. No. The proof takes place on a common test-function domain. A spectral conclusion also requires control of operator domains and of the evolution or differentiability of the relevant eigenfunctions. Those analytic conclusions do not follow from the algebra alone.